

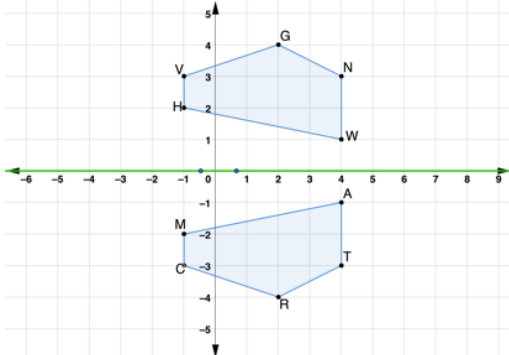
4.2 Algebraic Descriptions for Reflections

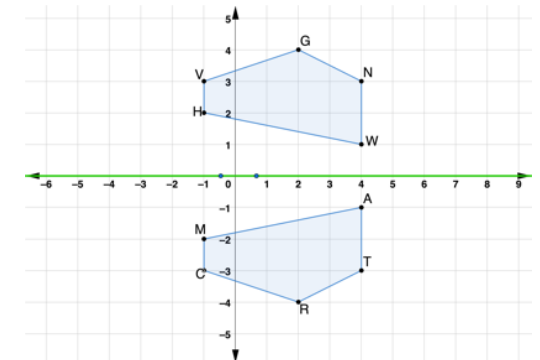
Lesson Introduction

 Benchmark	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.		 Learning Target	I can relate a reflection on the coordinate plane to an algebraic notation of the transformation.
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A	
 Essential Questions	- What is a reflection? - How can reflections be represented algebraically?			
 Lesson Narrative	The main part of the lesson, 4.2.2 Exploration, immerses students in an in-depth exploratory discovery of the effects of reflections across a variety of lines. Students focus on the ordered pairs of the vertices and the midpoint of line segments that connect corresponding vertices across the line of reflection. Students engage in language routines and math thinking and reasoning to draw conclusions leading toward generating algebraic descriptions of reflections.			
 Component Summary	4.2.1 Warm-Up (~5 min.)	Critical thinking problem (SE p. 184)	4.2.3 Bring It Together (5-10 min.)	Students summarize and synthesize their findings from the Exploration (SE p. 189)
	4.2.2 Exploration (25-30 min.)	Exploration of reflections across a variety of lines of reflection (SE p. 185)	4.2.4 Wrap-Up (~5 min.)	Students relate elements of reflections to a squish painting (SE p. 189)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Integer chips or other chips for 4.2.1 Warm-Up (Optional) Squish painting	N/A

 Teacher Tips	Common Misconceptions - Students may struggle differentiating between axes, values in an ordered pair, and their opposites. - Students who struggle in spatial reasoning might find it difficult to navigate the visual elements of reflections.	Things to Consider Consider using sentence stems to actively convert student verbal descriptions into mathematical language. For example, when students are describing the effects of reflection like, “They are in the same place,” provide scaffolded phrases like, “Vertex __ is the same distance away from __ as Vertex __.”
	Literacy and Language Routines Notice and Wonder 4.2.2 Exploration questions 1c-f provide two opportunities for students to make sense of their observations in mathematically meaningful ways. Encourage students to utilize Notice and Wonder techniques for each element to help each other discover and produce algebraic descriptions based on the Exploration.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 4.2.2 Exploration question 2b provides an opportunity for students to engage in mathematical discussion by analyzing and critiquing the thinking of others. Use this as an opportunity to provide students with practice analyzing the thinking of others and then articulating their own thinking in response.
	 Differentiate Instruction For a kinesthetic entry point, consider using physical manipulatives such as patty paper or cut-outs for students to explore the reflections in 4.2.2 Exploration.	

Lesson Notes:





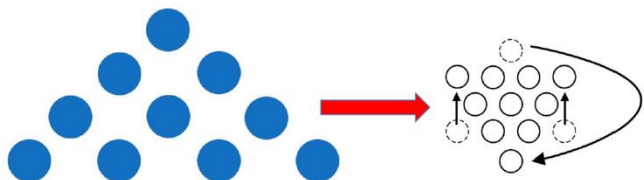
4.2.1 Warm-Up: Moving Circles

Component Overview: Students engage with a puzzle to determine how to solve it by moving only three circles.



4.2.1 Warm-Up: Moving Circles

1. Move any three circles so that the pyramid points down.



The arrows provided here are for answer key purposes. Students will only have the blue dots visible.

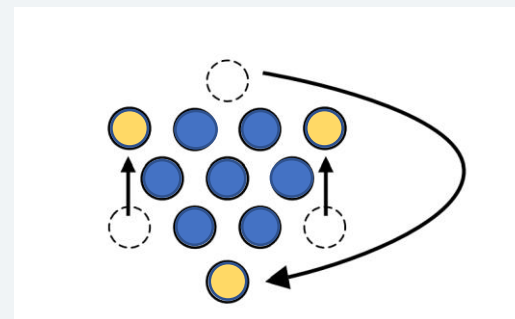


Consider prompting questions to target specific actions of Growth Mindset.

- Was the puzzle a struggle, or was it easy for you?
- What did you look for first when making pyramid point down?
- What did you try first? What strategy or different way of looking at the pyramid helped you know what to do next?



Do you have integer chips or similar manipulatives? Students could interact with this pyramid prompt for a kinesthetic learning experience. This could also be done with digital manipulatives.



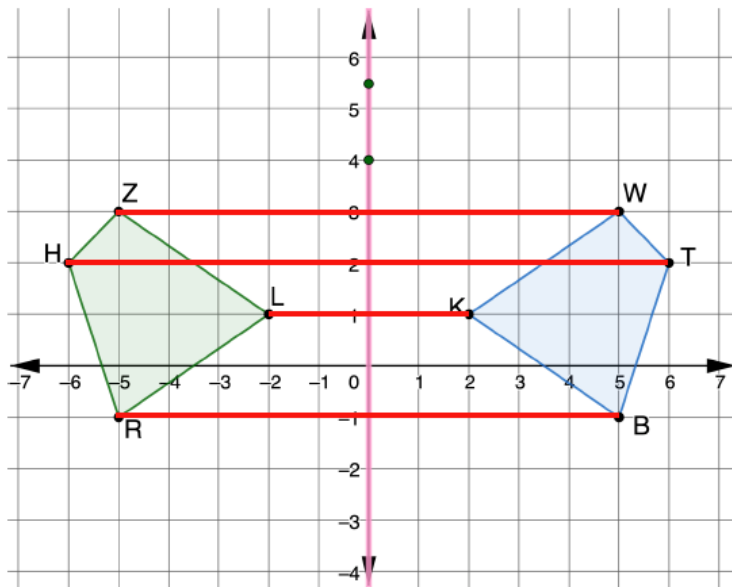
4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections

Component Overview: Students will explore how changes in x -values and y -values of ordered pairs of corresponding vertices after a reflection can be used to determine algebraic descriptions of the reflections. Students explore reflections across the y -axis, x -axis, the origin, the line $y = x$, and the line $y = -x$.



4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections

1. Quadrilaterals $HZLR$ and $TWKB$ are shown on the coordinate grid. Quadrilateral $HZLR$ was reflected across the y -axis to create quadrilateral $TWKB$.



- a. Draw a line segment between each set of corresponding vertices.
Shown on figure.
- b. Describe how the line segments between the vertices are related to the y -axis.
Answers will vary. Each vertex is the same distance from the y -axis as its corresponding vertex. The y -axis bisects each line segment. The line segments between the vertices are perpendicular to the y -axis.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.



During the exploration, allow students to describe relationships using their own words first. Students informally experienced reflections in Unit 1, so the exploration helps them recall that information. For students who are struggling with descriptions, refer them to Unit 1. As you circulate, if student descriptions show misunderstanding, then ask probing questions like:

- What is the relationship $\overline{ZW}$ has with the y -axis?
- Is that relationship the same for the other line segments?

4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

c. Complete the table.

	<i>H to T</i>	<i>Z to W</i>	<i>L to K</i>	<i>R to B</i>
Distance	12	10	4	10
Ordered Pair of the Midpoint	(0,2)	(0,3)	(0,1)	(0,-1)

d. What do you notice about the midpoints?

Answers will vary. All x -values are zero. All of the midpoints are on the y -axis.

e. The table shows the coordinates of each vertex.

Vertex	x	y	Vertex	x	y
<i>H</i>	-6	2	<i>T</i>	6	2
<i>Z</i>	-5	3	<i>W</i>	5	3
<i>L</i>	-2	1	<i>K</i>	2	1
<i>R</i>	-5	-1	<i>B</i>	5	-1

Work with your partner to describe how the coordinates of the pre-image, quadrilateral $HZLR$, are related to the coordinates of the image, quadrilateral $TWKB$.

Answers will vary. The x -values are opposites and the y -values are the same.

f. Write a conjecture that you and your partner think will work for the algebraic description of a reflection.

Answers will vary. Since the polygon is reflecting over the y -axis, the x -values change to their opposite sign.

$(x, y) \rightarrow (-x, y)$

g. Discuss with your partner if the algebraic description will work for any reflection. Summarize your conclusion.

Answers will vary. The point or line of reflection matters. Reflections over the y -axis will follow that description, but if the line of reflection is not the y -axis, then a new algebraic description would be needed.



Notice and Wonder Routine

Notice the intentional scaffolding of language routine elements embedded within the Exploration. Questions 1c-1d and 1e-1f provide the space and opportunity for students to discover how the ordered pairs of the vertices after a reflection can be used to generate algebraic descriptions. Students may not be engaging in a formal Notice and Wonder routine, but they can be prompted to notice and wonder about what they see for additional entry points when stuck in parts like question 1e.

Emphasize the word “opposite” in student discussion, as this word is seen in later questions of this lesson.

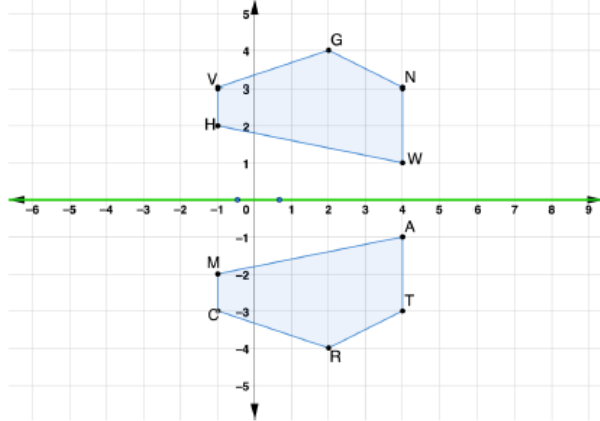


MA.K12.MTR.5.1: Patterns and Structure

The exploration provides students the opportunity to use patterns and structure to help understand and connect previously learned concepts (reflections) to new concepts (representing them with algebraic descriptions).

4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

2. Polygons $MCRTA$ and $HVGNW$ are shown on the coordinate grid. Polygon $MCRTA$ was reflected across the x -axis to create polygon $HVGNW$.



- a. Describe the relationship that the line segments between each set of corresponding vertices have with the x -axis.

Answers will vary. Each vertex is the same distance from the x -axis as its corresponding vertex. The line segments between corresponding vertices are perpendicular to the x -axis.

- b. Wendall and Samah were working together on writing an algebraic description for reflections across the x -axis. Each student's algebraic description and explanation of their thinking is shown in the table.

Wendall	Samah
$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x, -y)$
Because the polygon flipped over the x -axis, the x becomes its opposite.	Because above the x -axis all y -values are positive and below the x -axis all y -values are negative, the y becomes its opposite.



Monitor student discussion for question 2a. If necessary, prompt students to use the scaffolding from question 1 similarly in question 2. Draw line segments between each set of corresponding vertices from polygon $MCRTA$ to polygon $HVGNW$.

Prompting questions could include the following:

- What do you notice about the midpoints of the line segments?
- What do you notice about the coordinates of each ordered pair?
- What do you notice about the line of reflection and the line segments?
- How are these relationships similar to and different from the previous reflection?



MA.K12.MTR.4.1: Discussion and Reflection

Students have the opportunity to critique the reasoning and thinking of others. Consider having students or their partners read this passage out loud for others to listen and respond to in conversation. The error analysis of question 2b has students look at the common misconception of a reflection across the x -axis resulting in the x -value changing.

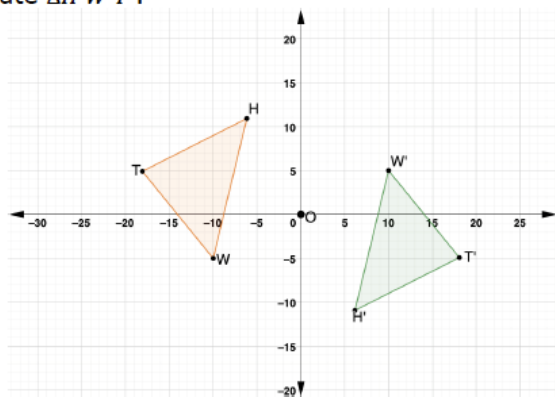
4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

Discuss with your partner which, if any, algebraic description is correct. Use at least two ordered pairs from polygons $MCRTA$ and $HVGNW$ to defend your choice. Write a summary of your discussion.

Samah is correct.

Answers will vary. Samah is correct because $A(4, -1)$ becomes $W(4, 1)$ and $R(2, -4)$ becomes $G(2, 4)$. You can see that the x -values remained the same but the y -values are now the opposite sign.

3. Triangles HWT and $H'W'T'$ are shown on the coordinate grid. Triangle HWT was reflected across the origin to create $\triangle H'W'T'$.



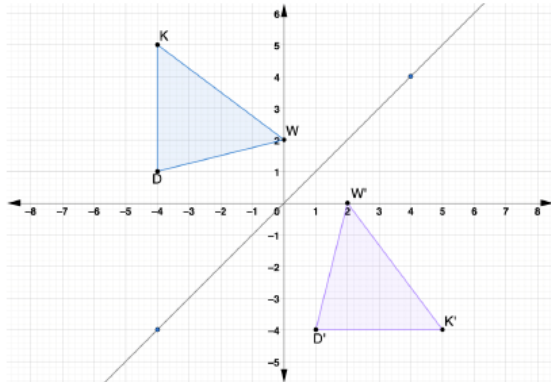
- Discuss with your partner the algebraic description for a reflection across the origin.
 $(x, y) \rightarrow (-x, -y)$
- Re-examine the algebraic descriptions of rotations. Which rotation is similar to a reflection across the origin?
 180° rotation



Question 3b seeks to build a conceptual connection to the previous lesson. Consider how to best engage students with the two elements of question 3 by beginning with a question like, “Does this look familiar? Why or why not?” and suggest they start with question 3b before finishing on question 3a.

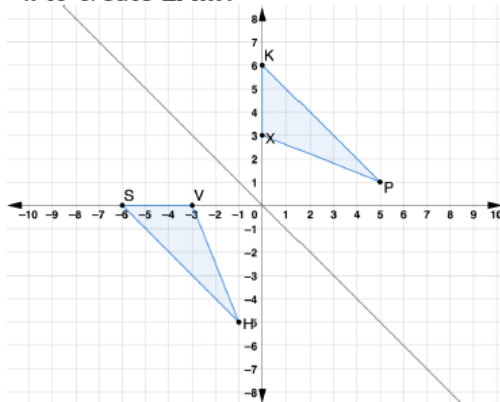
4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

4. Triangles KWD and $K'W'D'$ are shown on the coordinate grid. Triangle KWD was reflected across the line $y = x$ to create $\triangle K'W'D'$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = x$.
 $(x, y) \rightarrow (y, x)$

5. Triangles HSV and PKX are shown on the coordinate grid. Triangle HSV was reflected across the line $y = -x$ to create $\triangle PKX$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = -x$.
 $(x, y) \rightarrow (-y, -x)$



If time permits, transform your classroom into a coordinate plane and challenge students to create a mathematically accurate reflection using elements like desks, chairs, bags, and more!

4.2.3 Bring It Together: Algebraic Descriptions for Reflections

Component Overview: Students summarize and synthesize what they have discovered about reflections across both axes, $y = x$, and $y = -x$, including representing each by its algebraic description.



4.2.3 Bring It Together: Algebraic Descriptions for Reflections

Reflections	
Across the x -axis	Across the y -axis
The y -coordinate changes to its <u>opposite</u> . $(x, y) \rightarrow (x, -y)$	The x -coordinate changes to its <u>opposite</u> . $(x, y) \rightarrow (-x, y)$
Across $y = x$	Across $y = -x$
The coordinates <u>switch places</u> . $(x, y) \rightarrow (y, x)$	The coordinates <u>switch places</u> and become <u>opposite</u> . $(x, y) \rightarrow (-y, -x)$



Use this section to encourage synthesis, not memorization. Call on students to share their ideas following the exploration to highlight student voice and solidify takeaways from the Exploration.

4.2.4 Wrap-Up: Cool Down

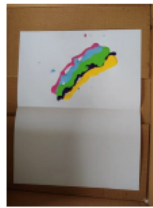
Component Overview: Students apply mathematical thinking and reasoning learned throughout the lesson to connect a line of reflection to art.



4.2.4 Wrap-Up: Cool Down

A reflection can be thought of as folding or "flipping" an object over the line of reflection. A squish painting is a fun application of a line of reflection.

To make a squish painting, a large sheet of construction paper is folded in half then opened up. Paint is dotted along one side of the fold. The paper is folded, and the folded paper is rubbed, pounded, and squished to move the paint around. The paper is then opened up to reveal the squish painting.



1. How does a squish painting show that in a reflection, each point of the original figure and its image are the same distance away from the line of reflection?

Answers will vary. In a squish painting, you can see how close each part of the painting is to the crease in the paper. You can see that the picture is a mirror image. The fold is the line of reflection.



Consider what this Cool Down might look like as a performance task or interdisciplinary project with an art class. Students could be immersed into an exploration of reflections in art to establish a strong foundation for key elements of reflections, while the 4.2.2 Exploration can serve to quantify and make mathematical sense of their findings of reflections in art.